Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

Chembur, Mumbai-400088

(An Autonomous Institute Affiliated to University of Mumbai)



Department of Cyber Security

presentation
on

The Role of Artificial Intelligence in Advanced Digital Forensics

(Subject:Advanced Digital Forensics HNCSCR0701)

by

(Priya Gaglani 23UF18027CM016)

Email: priya.18027@sakec.ac.in

Vision & Mission

Institute Vision

To become a globally recognized institution offering quality education and enhancing professional standards

Department Vision

To be a high repute focused department preparing students ethically in field of cyber security and make them capable to lead technical, economic, social and ethical development of the society.

Institute Mission

To impart high quality technical education to the students by providing an excellent academic environment, well - equipped laboratories and training through motivated teachers.

Department Mission

1. To provide an academic environment for the development by providing training and skill development to build knowledge, skills, values, and confidence to take a leadership role in the field of Cyber Security.
2. To cultivate research and entrepreneurship culture resulting in knowledge and innovative technologies that contribute to sustainable development of the society.
3. To enhance academic and corporate collaborations.

Program Specific Outcomes

Index	Outcome
PSO1	The Cyber Security graduates are able to gain a thorough understanding of the Cyber Security landscape with its growing threats and vulnerabilities in the world of computing including software and hardware.
PSO2	The Cyber Security graduates are able to attain skills to comprehend and anticipate future challenges and devise methods to meet them and also, be articulate and skilled to convince all the stakeholders
PSO3	The Cyber Security graduates are able to acquire and demonstrate the ability to use ethical standard tools, practices and technologies for the analysis, design, development, implementation and testing of innovative and optimal Cyber Security solutions without compromising the privacy needs of individual and entities and the security concerns of law enforcement agencies.

Outline

- Introduction
- Problem Statement & Objectives
- Core Concept / Theory
- Working / Mechanism
- Functional & Non-Functional Requirements (if applicable)
- Challenges / Limitations
- Case Study / Example
- Comparative Analysis (if applicable)
- Tools & Modeling Approaches
- Applications in Real Life
- Proposed Framework / Design
- Conclusion & Future Scope
- Reference

Introduction & Objectives

Digital forensics process

Identify	Collect	Preserve	**Examine**	Present

AI can assist at this stage

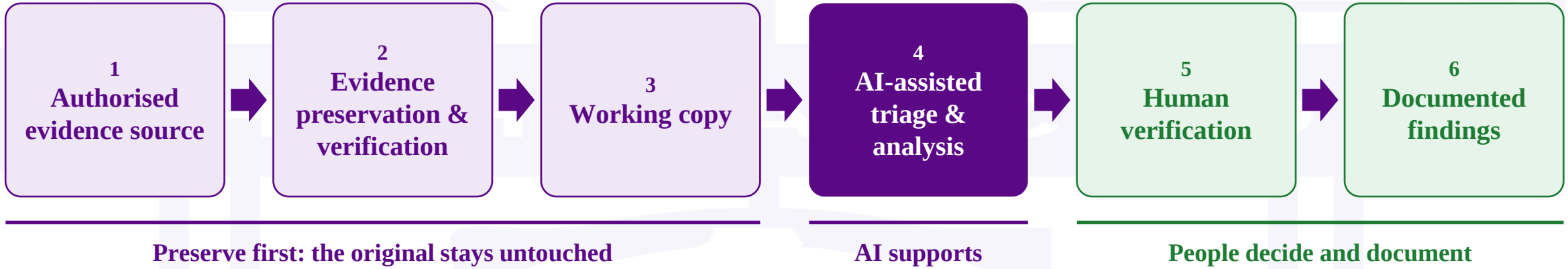
Problem statement

- Phones, laptops and cloud accounts hold far more data than an investigator can read by hand.
- AI tools can help sort and flag this data, but an unverified AI result can mislead an investigation or weaken the evidence.

Objectives

1. Show where AI supports forensic work
2. Present a workflow that protects evidence
3. Demonstrate rule-based flagging on synthetic data
4. Explain why every AI result needs human verification

AI-Assisted Forensic Workflow



AI works on the copy

Analysis runs on a verified working copy, never on the original evidence.

AI output is a lead

Results enter the workflow at step 5 and must be checked before they count.

Everything is recorded

Tool version, settings and steps are documented so others can review the work.

Where AI Helps Investigators

1 Data triage and classification

Helps: Prioritises and groups large file sets and flags likely duplicates.

Check: Confirm the grouping is relevant before relying on it.

2 Detecting unusual activity

Helps: Flags odd login times, file access or data transfers in logs.

Check: An unusual event can have an innocent explanation.

3 Text and multimedia analysis

Helps: Searches messages and documents; speech-to-text and image tools speed up review.

Check: Unclear audio, slang or poor images cause errors. Compare with originals.

4 Connecting evidence

Helps: Relates timestamps, accounts and files across devices and logs to build a timeline.

Check: Similar timestamps do not prove two events are linked.

Synthetic Login-Log Scenario

HYPOTHETICAL DATA: not a real case

Time	User	Source IP	Result
02:41:12	user_d	198.51.100.7	SUCCESS
02:41:50	user_c	203.0.113.5	FAILED
02:42:03	user_c	203.0.113.5	FAILED
02:42:17	user_c	203.0.113.5	FAILED
09:12:05	user_a	192.0.2.10	SUCCESS
09:45:30	user_b	192.0.2.11	SUCCESS
10:03:18	user_c	192.0.2.12	SUCCESS
11:20:44	user_b	192.0.2.11	SUCCESS
14:05:09	user_a	192.0.2.10	SUCCESS
16:30:00	user_a	192.0.2.10	SUCCESS

Shaded rows = flagged by the baseline rules on the next slide

Baseline we defined

- Normal hours: 08:00 to 17:59
- Flag a failed login if the same user has 3 or more failures
- *IPs are reserved documentation addresses*

Unusual is not unauthorised

A flag only says: look here first. A night-shift admin or a user in another time zone could produce the same pattern.

Rule-Based Flagging in Python

```
from collections import Counter

# logs = the 10 records from the previous slide
NORMAL_START, NORMAL_END = 8, 18
MAX_FAILURES = 3

failed = Counter(user for t, user, ip, result in logs
                 if result == "FAILED")

flagged = 0
for t, user, ip, result in logs:
    reasons = []
    hour = int(t[:2])
    if hour < NORMAL_START or hour >= NORMAL_END:
        reasons.append("OFF_HOURS")
    if result == "FAILED" and failed[user] >= MAX_FAILURES:
        reasons.append("REPEATED_FAILS")
    if reasons:
        flagged += 1
        print(t, user, ip, result, "|", ", ".join(reasons))

print(flagged, "of", len(logs), "events flagged for human review")
```

In plain English

Baseline: NORMAL_START, NORMAL_END and MAX_FAILURES are the rules we set in advance.

Counter: counts how many failed logins each user has.

The loop: checks every record against both rules and collects the reasons.

Output: only flagged rows are printed, for a person to review.

Rule-based flagging, not machine learning.
A teaching example, not a forensic tool.

From Alert to Conclusion

Output (script run and checked)

```
02:41:12 user_d 198.51.100.7 SUCCESS | OFF_HOURS
02:41:50 user_c 203.0.113.5 FAILED | OFF_HOURS, REPEATED_FAILS
02:42:03 user_c 203.0.113.5 FAILED | OFF_HOURS, REPEATED_FAILS
02:42:17 user_c 203.0.113.5 FAILED | OFF_HOURS, REPEATED_FAILS
4 of 10 events flagged for human review
```

Possible innocent reasons

- On-call or night-shift work
- Staff working from another location
- A user who forgot a password

What adds concern

- Three failures in about 30 seconds
- An IP not seen elsewhere in the log
- Needs checking: user_c logs in at 10:03 from a different IP

From alert to conclusion

AI or rule alert
What the tool flags

↓

Investigative lead
Worth a closer look

↓

Verify
Check original evidence and context

↓

Documented conclusion
Written, explained, reviewable

Evidence Integrity: Hash Check

Original evidence `login_log_v1`	SHA-256 →	`1b7db09ec731cc83…`	**Reference value, recorded at acquisition**
Working copy A `login_log_v1`	SHA-256 →	`1b7db09ec731cc83…`	**MATCH: contents identical to the reference**
Working copy B `login_log_v2`	SHA-256 →	`b83c2b08ff4b5af2…`	**MISMATCH: investigate before using**

Changing one character gives a completely different hash. Only the first 16 of 64 hex characters are shown.

A matching hash shows the contents are unchanged. It does not prove the original was authentic or that chain of custody was kept.

Benefits, Risks & Safeguards

Manages large data volumes

Surfaces patterns that are easy to miss

Applies the same process consistently

Concern	Why it matters	Safeguard
False positives / negatives	Harmless events get flagged; relevant evidence can be missed	Verify key findings against source evidence; use a second method where practical
Privacy	Devices hold personal data unrelated to the case	Stay within legal authorisation; limit access to sensitive data
Explainability	Some tools cannot show why they flagged an item	Use tools that log operations; explain the method in the report
Evidence integrity	Careless analysis could alter or weaken evidence	Work on verified copies; keep hashes and custody records
Bias and manipulation	Weak results on unfamiliar data; misleading inputs	Test and validate tools; keep human review
Human oversight	AI cannot establish that a crime occurred	Investigator interprets, decides and documents tool, settings and steps

Future Scope & Conclusion

Future scope

- Evidence is spreading across phones, cloud services, connected devices and networks, so AI will help compare sources.
- More automation means transparent methods, independent validation and careful data handling matter more.
- Investigators still need technical knowledge, critical thinking and judgement.

Conclusion

- AI helps organise data, flag unusual activity, analyse text and media, and suggest links between events.
- Its output must be verified, and evidence must be preserved and documented.

An AI alert is a lead, not a conclusion.

References

[1] Priya. Gaglani, “The Role of Artificial Intelligence in Advanced Digital Forensics,” ADF SLA 1 article. (primary source).

[2] K. Kent, S. Chevalier, T. Grance and H. Dang, Guide to Integrating Forensic Techniques into Incident Response, NIST SP 800-86, Aug. 2006. https://doi.org/10.6028/NIST.SP.800-86

[3] B. Guttman, D. R. White, S. Williams and T. Walraven, Digital Evidence Preservation: Considerations for Evidence Handlers, NIST IR 8387, Sep. 2022. https://doi.org/10.6028/NIST.IR.8387

[4] NIST, Secure Hash Standard (SHS), FIPS 180-4, Aug. 2015. https://doi.org/10.6028/NIST.FIPS.180-4

[5] Python Software Foundation, “hashlib” and “collections.Counter,” Python 3 documentation. https://docs.python.org/3/library/hashlib.html

[6] J. Arkko, M. Cotton and L. Vegoda, IPv4 Address Blocks Reserved for Documentation, RFC 5737, Jan. 2010. https://www.rfc-editor.org/rfc/rfc5737

Slides 5–7 and 12–13 follow my SLA 1 article [1]. Slides 8–11 (scenario, code, hash example) are original demonstration material on synthetic data, supported by [2]–[6].

Thank You